

Pre-registration: Independent Monte Carlo Verification of Paleopole Orientation Clustering in the Buildreps Database

Salah-Eddin Gherbi

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Author: Salah-Eddin Gherbi

Email: salahealer@gmail.com

ORCID: <https://orcid.org/0009-0005-4017-1095>

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1. Summary

This document pre-registers a statistical analysis to test, independently, the central empirical claim of Mario Buildreps’ “Archaeorientation” framework: that the orientations of ancient pyramids, temples, and megalithic structures cluster around five proposed former-pole positions along the $\sim 47^\circ\text{W}$ meridian. The analysis will be a Monte Carlo simulation comparing the observed clustering to a null distribution generated by shuffling folded orientations while preserving site geography. This document and its SHA-256 hash are deposited on Zenodo and committed to GitHub **before** the database is opened, so that the analysis methodology cannot be revised after seeing the data.

2. Background

Mario Buildreps has assembled a database of approximately 1,159 ancient structures worldwide, including their geographic coordinates and orientations relative to current true north. He proposes that 993 of these structures (those whose great-circle orientations intersect a meridian band around 47°W within the analyzable latitude range) cluster around five paleopole locations: (76.0°N , 47°W), (72.2°N , 47°W), (64.1°N , 47°W), (52.3°N , 47°W), and the current geographic pole at (90°N), with a sixth “out-of-scope” candidate pole at (42.0°N , 47°W). His published probability statements ($\sim 100\%$ certainty for poles I–III, $\sim 99.999\%$ for IV–V) are based on a binomial test against a uniform null distribution along the 47°W meridian, with non-uniform “Dynamical Grouping” of latitude bins.

Two methodological concerns motivate independent re-analysis. First, a uniform-along-meridian null does not account for the geographic distribution of sites, which alone produces structure in the distribution of great-circle intersection latitudes. Second, the published bin widths and positions are non-uniform and appear to have been chosen post-hoc, which inflates apparent significance. A Monte Carlo against a geography-preserving null, with bin-free test statistics, is the appropriate test of the underlying claim.

3. Scope and limits

This pre-registration tests one narrow empirical claim:

The folded northernface orientations of ancient structures in the Buildreps database point toward the five proposed pole positions (and the candidate sixth) more than expected under random orientations applied to the same geographic distribution of sites.

This pre-registration does not test, support, or refute any of the broader interpretive claims associated with the framework, including but not limited to: Earth expansion, crustal-deformation theories of ice ages, the antiquity of ancient structures beyond conventional chronology, the cause of the Greenland ice sheet, or any model of human prehistory. The outcome of the orientation test is logically independent of those questions and is not evidence for or against them.

4. Data

File Database_Mario_Buildreps_V14.xlsx

Source Provided by Mario Buildreps via email on the date recorded in the repository commit history, under the explicit condition that the raw data not be redistributed.

SHA-256 hash <actual-hash-string-here>

Hash provenance The hash was computed, GPG-signed, and OpenTimestamped on the Bitcoin blockchain on 17 May 2026, recorded in the repository at commit 426dd95f4f1d62dbb2ea6b7be0bd2d1499834fb8b2c923ca59299384fd4ddb7c, **before this pre-registration was written.**

The data file is not redistributed. Anyone obtaining the database from the source can verify byte-for-byte equality with the file analyzed here by computing the SHA-256 hash and comparing it to the recorded value.

Per the rules document provided by the data owner (“BASIC RULES FOR PROBABILITIES AND FINDING MOST PROBABLE POLE POSITIONS”):

- The database contains roughly 1,159 structures total; 993 are considered in-range (folded orientation in $[-45^\circ, +45^\circ]$ with great-circle intersection on the 47°W meridian) and 166 are out of range (“No Intersect 47.1W ”).
- Orientations are recorded as folded northernface azimuths.
- Multiple similarly oriented structures at a single site (within approximately 2° , per measurement error) are averaged to a single entry.

5. Hypotheses

Null hypothesis (H_0): The folded orientations of the 993 in-range structures are statistically independent of site geography in the following operational sense: assigning each structure a folded orientation drawn at random (with replacement) from the empirical distribution of folded orientations in the database produces a clustering of great-circle pole-pointing as strong as observed.

Alternative hypothesis (H_1): The folded orientations point toward the proposed pole positions more closely than expected under H_0 .

This is a one-sided test.

6. Primary test statistic

For each in-range structure i with location (φ_i, λ_i) and folded northernface azimuth $\alpha_i \in [-45^\circ, +45^\circ]$, I will:

1. Compute the great circle defined by extending the structure’s northernface direction forward over the sphere.
2. Find the latitude φ'_i at which this great circle crosses the 47°W meridian.
3. For each of the five proposed poles $k \in \{\text{I, II, III, IV, V}\}$ at latitudes $\varphi_k \in \{90^\circ, 76.0^\circ, 72.2^\circ, 64.1^\circ, 52.3^\circ\}$, compute the absolute angular difference along the meridian:

$$d_{i,k} = |\varphi'_i - \varphi_k|$$

4. Take the minimum over the five poles:

$$d_i = \min_k d_{i,k}$$

The **primary test statistic** is:

$$T = \frac{1}{N} \sum_{i=1}^N d_i$$

where $N = 993$ is the number of in-range structures. Smaller T means orientations point on average more closely to the nearest of the five proposed poles. Under H_0 , T takes a value determined by the geography of sites and the empirical orientation distribution. Under H_1 , T is smaller than expected.

This test statistic is **binning-free** and does not depend on any post-hoc choices of bin width or location.

7. Null distribution and Monte Carlo procedure

I will generate the null distribution of T as follows:

1. Read the 993 in-range structures' coordinates (ϕ_i, λ_i) from the database. **These remain fixed across all iterations.**
2. Read the 993 in-range folded orientations and store them as a vector A .
3. For each Monte Carlo iteration $m \in \{1, \dots, M\}$:
 - a. Randomly permute A to produce $A^{(m)}$, so that each site retains its location but receives a randomly drawn folded orientation from the empirical pool (sampling without replacement, preserving the marginal distribution).
 - b. Compute $T^{(m)}$ using the procedure in §6.
4. Repeat for $M = 10,000$ iterations.

The **p-value** for the primary test is:

$$p = \frac{1 + \#\{m : T^{(m)} \leq T_{\text{obs}}\}}{1 + M}$$

I will reject H_0 at the $\alpha = 0.05$ level if $p < 0.05$.

This null preserves: (a) the geographic distribution of sites exactly, and (b) the marginal distribution of folded orientations exactly. It tests specifically whether the *pairing* of orientations to locations is consistent with random assignment.

8. Pole VI handling

The candidate Pole VI at (42.0°N, 47°W) is described by the data owner as “out of scope” because including it suppresses the probabilities of the five primary poles. Because the original published claim specifies five poles, **the primary confirmatory test will use only Poles I-V**. The inclusion of a sixth pole is run as a **sensitivity**

analysis, pre-specified here, to assess whether the exclusion of Pole VI by the data owner could constitute selection on outcome.

- **Test 8a (primary)**: with five poles (I-V).
- **Test 8b (sensitivity)**: identical procedure, with six poles (I-VI).

The primary inference of this pre-registration will be based on Test 8a. Test 8b is pre-registered, will be run regardless of the outcome of Test 8a, and will be reported transparently as a sensitivity analysis — not as a post-hoc exploration.

9. Significance threshold, multiple comparisons, and reporting conventions

The primary test uses a single test statistic T , so no multiple-comparison correction is required for the primary result. The primary significance threshold is $\alpha = 0.05$, **one-sided**.

For the per-pole secondary analysis (§11), I will use the Šidák correction for five (or six, in the sensitivity analysis) simultaneous tests at the family-wise $\alpha = 0.05$ level.

Reporting conventions:

- Exact p-values will be reported throughout (e.g., “ $p = 0.034$ ”), not only the conclusion relative to a threshold. The conventional thresholds are reporting aids, not the substantive content.
- A result with $p < 0.01$ will be reported as “highly significant”; $0.01 \leq p < 0.05$ as “significant”; $p \geq 0.05$ as “not significant at the pre-registered threshold.”
- For Monte Carlo p-values that come out as exactly 0 in finite simulations, the upper bound $1/(M+1)$ will be reported (e.g., “ $p < 0.0001$ ($M = 10,000$)”).

10. Look-elsewhere control (longitude scan)

The choice of 47°W as the reference meridian is itself a researcher degree of freedom, since the meridian was identified by the data owner as the longitude of strongest clustering in his exploratory analysis. To address this, I will perform a two-step longitude scan:

Step 10a (primary scan, 5° resolution). For each candidate longitude $\lambda_c \in \{-180^\circ, -175^\circ, \dots, +175^\circ\}$ (72 longitudes at 5° spacing), I will compute $T(\lambda_c)$ using the procedure in §6 with the meridian relocated to λ_c . I will then compare $T_{\text{obs}}(47^\circ\text{W})$ to the empirical distribution of $T_{\text{min}} = \min_{\lambda_c} T(\lambda_c)$ across the same M Monte Carlo iterations.

The look-elsewhere-corrected p-value is:

$$p_{\text{LEE}} = \frac{1 + \#\{m : T_{\text{min}}^{(m)} \leq T_{\text{obs}}(47^\circ\text{W})\}}{1 + M}$$

The 5° -resolution p_{LEE} is the **primary look-elsewhere outcome**.

Step 10b (sensitivity, 1° resolution, conditional). If p_{LEE} at 5° resolution is below 0.05, I will re-run the scan at 1° resolution (360 longitudes) as a higher-resolution sensitivity analysis. The 1° scan is pre-registered here as a conditional follow-up — its execution depends on the 5° outcome, but its specification is locked in by this document. The 1° result will be reported alongside the 5° primary result; only the 5° result counts toward the primary inference.

11. Secondary and sensitivity analyses

All pre-specified, all to be reported regardless of outcome.

(a) Per-pole confirmatory test. For each of the five (or six) poles individually, compute the count of structures whose great-circle intersection on the 47°W meridian falls within $\pm 1.5^\circ$ of φ_k . Compare to the null distribution of the same count under permuted orientations. Report Šidák-corrected p-values.

(b) Site-to-pole assignment test. The data owner publishes an explicit assignment of named sites to specific poles (e.g., Giza → Pole I, Babylon → Pole II, Teotihuacan → Pole III). For sites in the database that match the published list, test whether each site’s orientation points to its assigned pole within $\pm 1.5^\circ$ more often than expected under the null. This is a stronger confirmatory test of the framework’s specific predictions.

(c) Aggregation-threshold sensitivity. Repeat the primary test with the per-site averaging threshold varied across $\pm 1^\circ$, $\pm 2^\circ$, and $\pm 3^\circ$. If the threshold is not explicit in the data, I will re-aggregate the relevant sites from the raw data.

(d) Geographic-block null. Replace the global orientation shuffle with a block-permutation null that shuffles orientations only within geographic/cultural regions (Americas, Europe-Mediterranean, Middle East, South Asia, East Asia, Africa, Oceania, defined by simple longitude/latitude bands specified in the analysis code). This tests robustness to potential within-region orientation correlation.

(e) Both-axes test. A rectangular structure has 90° rotational symmetry; each contributes two orthogonal axes that could in principle point at a pole. The primary test uses only the northernface axis (matching the data owner’s convention). Repeat the primary test using both axes per structure and report.

(f) Bin-based test mirroring the original analysis. As an explicit point of comparison, repeat the per-pole counts using fixed 3°-wide bins centered on each pole latitude (the cluster width described in the data owner’s rules), and report under both the uniform-binomial null he uses and the geography-preserving Monte Carlo null specified here.

12. Pre-committed reporting

I commit to the following regardless of the outcome:

1. **All pre-registered tests will be reported**, including those that do not reach significance and those that contradict the data owner’s published probability claims.

2. **The data owner will receive the results 14 days before public release.** This window is for the data owner to identify factual errors and to provide his own commentary, which I will include alongside the analysis. Publication is not conditional on his agreement with the conclusions; the 14-day notice is a courtesy and a check against misrepresentation, not a veto.
3. **No new tests will be added post-hoc without clear labeling.** Any analysis run after the database is opened that was not pre-registered here will be labeled explicitly as exploratory in the final write-up and will not be used to make confirmatory claims.
4. **Method changes after seeing the data are prohibited.** If a clear specification error in this pre-registration is discovered after opening the data (e.g., an impossible computation), I will document the error, the correction, and the reason in the final write-up, alongside the originally pre-registered analysis where possible.
5. **The raw database will not be redistributed.** Only summary statistics, code, figures, and the conclusions will be made public.
6. **Exact p-values will be reported, not only threshold conclusions.** See §9.

13. Code and reproducibility

All analysis code will be written in Python and committed to the `analysis/` directory of the repository. The code will:

- Read `Database_Mario_Buildreps_V14.xlsx` from a local, gitignored `data/` directory.
- Verify the SHA-256 hash matches the value recorded in this pre-registration before running. If the hash does not match, the script will exit with an error.
- Use a fixed pseudo-random seed (specified in the script) so that Monte Carlo results are bit-for-bit reproducible.
- Produce all figures and summary statistics in `results/`.

The Python environment will be pinned via `requirements.txt`. A reader who obtains the database from the source can reproduce the analysis exactly.

14. Provenance and timestamping

This pre-registration document will be:

1. Committed to the GitHub repository in `preregistration/` after being converted to PDF.
2. GPG-signed with the author's public key.
3. OpenTimestamped on the Bitcoin blockchain.
4. Deposited on Zenodo with a permanent DOI.

The Zenodo DOI will be added to this document as a footnote in the repository copy after deposit. The deposit will reference the GitHub repository and the SHA-256 hash of the database file.

15. Contact

Salah-Eddin Gherbi

Email: salahealer@gmail.com

ORCID: <https://orcid.org/0009-0005-4017-1095>

GitHub: <https://github.com/salahealer9>

Repository: <https://github.com/salahealer9/paleopole-orientation-verification>

This pre-registration is licensed under CC-BY-4.0. The associated analysis code, when committed, will be licensed under the MIT License. The database analyzed is not covered by either license and remains the property of Mario Buildreps.